

## Lab 3: Architectural Pattern Selection & Justification

Community Solar Credit Allocation Manager (Problem Statement #63) | Student: Hemant, PES University

### Architecture Selection

**We chose Layered Architecture for the Community Solar Credit Allocation Manager System.** The system is organised into a Presentation Layer (Web Portal UI), a Business Layer (Authentication Service, Credit Transfer Manager, Billing Offset Service, Telemetry Ingestion Service) and a Data Layer (Credit Ledger Database). Components talk only through the interfaces IAuth, ITransfer, IBilling, ITelemetry and IDataAccess shown in the component diagram.

| Style         | Fit for this scenario                                                                                                                                                                                   | Verdict         |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Layered       | Simple single deployment; clear UI / rules / data split; one shared ledger keeps balances consistent. Scaling single components is harder, which is acceptable at ~500 households.                      | <b>Selected</b> |
| Microservices | Independent scaling and fault isolation, but adds network latency, operational complexity and distributed-transaction problems for debit/credit transfers; unjustified for one neighbourhood microgrid. | Rejected        |
| Client-Server | Easy central control, but gives no internal structure for separating telemetry, transfer and billing logic, and the server is a single point of failure.                                                | Rejected        |

### Two Reasons for the Choice

**1. Scale and data consistency match the scenario.** The portal serves one neighbourhood microgrid (about 500 households, hourly readings), so transaction volume is modest. Peer-to-peer transfers must debit one prosumer and credit a neighbour atomically (FR-001). A single Credit Ledger Database behind the IDataAccess interface gives simple ACID transactions, whereas microservices with separate data stores would need distributed transactions.

**2. The three layers map directly to the domain's concerns.** Dashboards for the Prosumer Resident and Co-op Manager stay in the UI layer; transfer rules, billing offsets and meter-data validation stay in the business layer; storage stays in the data layer. Because each component hides its implementation behind an interface, one can be replaced without touching the others (for example, changing the utility's billing rules in the Billing Offset Service, or the meter protocol in the Telemetry Ingestion Service).

### Security Advantage

Only the business layer can reach the ledger, through IDataAccess, so the UI and external devices never touch stored data directly. The Authentication Service issues role-based session tokens (Prosumer Resident vs Co-op Manager) that the Credit Transfer Manager validates before every transfer, and balance and ownership checks live in one place, which blocks overdrafts and unauthorised transfers. Smart meter readings are validated at the Telemetry Ingestion Service before being stored, and all external interfaces (UI and meter feed) run over TLS.

### Performance Benefit

The Telemetry Ingestion Service is a separate component that writes hourly readings for all households in batches, so the ingestion load does not block interactive credit transfers. Calls between business components are in-process method calls rather than network hops, which keeps transfer latency low, and the data layer can use connection pooling and indexed balance lookups for fast queries. The trade-off is that the whole application scales as one unit, which is acceptable at this size.

### Component and Interface Summary

| Interface   | Provided by                 | Required by                      | Data flow / protocol                        |
|-------------|-----------------------------|----------------------------------|---------------------------------------------|
| IAuth       | Authentication Service      | Web Portal UI                    | login, session token (REST/HTTPS)           |
| ITransfer   | Credit Transfer Manager     | Web Portal UI                    | transfer request, balance query (REST-JSON) |
| IBilling    | Billing Offset Service      | Credit Transfer Manager          | apply credits to statement (internal API)   |
| ITelemetry  | Telemetry Ingestion Service | Smart Meter Feed                 | hourly kWh readings (MQTT/HTTPS)            |
| IDataAccess | Credit Ledger Database      | Transfer Mgr, Billing, Telemetry | SQL queries / writes (JDBC)                 |